MODEL LAB · 2026-10-19

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	24,115	3.04%	0.6080
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-06, scored on the 24,115 that came after (actual HR rate 3.04%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5898	1.67×	0.02937	0.13496	0.00609
KNN comps Prior is the pooled rate of the nearest comparables.	0.6080	1.57×	0.02932	0.13372	0.00385
SVR Support-vector regression on the same feature matrix.	0.6031	1.60×	0.02934	0.13424	0.00490
Random forest Forest regression; a learned similarity metric.	0.6048	1.60×	0.02932	0.13379	0.00336
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6080	1.57×	0.02932	0.13372	0.00385
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6080	1.57×	0.02932	0.13372	0.00385
Linear regression Ordinary least squares on the same features.	0.6065	1.60×	0.02934	0.13395	0.00406
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6059	1.64×	0.02936	0.13406	0.00525
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6051	1.60×	0.02933	0.13387	0.00397
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6041	1.58×	0.02933	0.13393	0.00438

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **258** knn **258** svr **258** rf **258** core **258** ensemble **258** linear **258** logistic **258** nn **258** xgb **258**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,467 PA, actual HR rate 2.39%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5828	1.52×	0.02336	0.11312	0.01660
KNN comps Prior is the pooled rate of the nearest comparables.	0.6179	2.20×	0.02327	0.11072	0.01033
SVR Support-vector regression on the same feature matrix.	0.5974	1.35×	0.02342	0.11447	0.01757
Random forest Forest regression; a learned similarity metric.	0.6043	1.69×	0.02335	0.11144	0.01396
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6179	2.20×	0.02327	0.11072	0.01033
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6179	2.20×	0.02327	0.11072	0.01033
Linear regression Ordinary least squares on the same features.	0.6177	1.69×	0.02349	0.11254	0.01413
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6177	1.35×	0.02353	0.11259	0.01721
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6138	1.86×	0.02330	0.11111	0.01206
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6084	1.52×	0.02339	0.11150	0.01594

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Junior Caminero 16.1%, Jo Adell 15.9%, Jonathan Aranda 13.7%, Patrick Bailey 13.0%, Yandy Díaz 12.3%, Chase DeLauter 11.9%
Flat league	6/6	Junior Caminero 15.4%, Jo Adell 15.2%, Jonathan Aranda 13.0%, Yandy Díaz 12.6%, Patrick Bailey 12.4%, Chase DeLauter 12.0%
KNN comps	6/6	Junior Caminero 15.8%, Jo Adell 15.7%, Jonathan Aranda 14.0%, Patrick Bailey 13.0%, Yandy Díaz 12.3%, Chase DeLauter 11.8%
SVR	6/6	Junior Caminero 17.3%, Jo Adell 16.2%, Jonathan Aranda 13.5%, Patrick Bailey 12.9%, Yandy Díaz 12.8%, Chase DeLauter 11.8%
Random forest	6/6	Junior Caminero 16.3%, Jo Adell 15.8%, Jonathan Aranda 13.5%, Patrick Bailey 12.7%, Yandy Díaz 12.7%, Chase DeLauter 11.7%
Core (KNN+SVR)	6/6	Junior Caminero 15.8%, Jo Adell 15.7%, Jonathan Aranda 14.0%, Patrick Bailey 13.0%, Yandy Díaz 12.3%, Chase DeLauter 11.8%
Forest blend	6/6	Junior Caminero 15.8%, Jo Adell 15.7%, Jonathan Aranda 14.0%, Patrick Bailey 13.0%, Yandy Díaz 12.3%, Chase DeLauter 11.8%
Linear regression	6/6	Junior Caminero 16.1%, Jo Adell 16.0%, Jonathan Aranda 13.7%, Patrick Bailey 12.9%, Yandy Díaz 12.5%, Chase DeLauter 11.8%
Logistic regression	6/6	Junior Caminero 16.5%, Jo Adell 16.3%, Jonathan Aranda 13.5%, Patrick Bailey 12.4%, Yandy Díaz 12.4%, Chase DeLauter 11.7%
Neural net (torch)	6/6	Junior Caminero 16.5%, Jo Adell 16.2%, Jonathan Aranda 13.6%, Patrick Bailey 12.7%, Yandy Díaz 12.3%, Chase DeLauter 11.6%
XGBoost	6/6	Junior Caminero 16.5%, Jo Adell 16.0%, Jonathan Aranda 13.7%, Yandy Díaz 12.9%, Patrick Bailey 12.8%, Chase DeLauter 11.7%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	Junior Caminero TB slate	16.1%	15.4%	15.8%	17.3%	16.3%	15.8%	15.8%	16.1%	16.5%	
2	Jo Adell CLE slate	15.9%	15.2%	15.7%	16.2%	15.8%	15.7%	15.7%	16.0%	16.3%	
3	Jonathan Aranda TB	13.7%	13.0%	14.0%	13.5%	13.5%	14.0%	14.0%	13.7%	13.5%	
4	Patrick Bailey CLE	13.0%	12.4%	13.0%	12.9%	12.7%	13.0%	13.0%	12.9%	12.4%	
5	Yandy Díaz TB	12.3%	12.6%	12.3%	12.8%	12.7%	12.3%	12.3%	12.5%	12.4%	
6	Chase DeLauter CLE	11.9%	12.0%	11.8%	11.8%	11.7%	11.8%	11.8%	11.8%	11.7%	
7	Travis Bazzana CLE	11.7%	11.7%	11.4%	11.3%	11.5%	11.4%	11.4%	11.5%	11.2%	
8	Nathaniel Lowe CLE	11.3%	11.1%	11.5%	11.3%	11.5%	11.5%	11.5%	11.3%	11.2%	
9	Victor Mesa Jr. TB	11.1%	10.9%	11.2%	11.7%	11.7%	11.2%	11.2%	11.4%	11.4%	
10	José Ramírez CLE	10.2%	10.5%	10.3%	10.3%	10.4%	10.3%	10.3%	10.6%	10.9%	
11	Petey Halpin CLE	9.3%	10.3%	9.3%	9.6%	9.4%	9.3%	9.3%	10.3%	9.8%	
12	Jonny DeLuca TB	8.1%	8.6%	7.8%	7.7%	7.3%	7.8%	7.8%	7.8%	8.0%	
13	Liam Hicks TB	7.6%	8.6%	7.4%	8.2%	7.3%	7.4%	7.4%	7.9%	7.8%	
14	Brayan Rocchio CLE	7.2%	7.7%	7.1%	7.1%	6.8%	7.1%	7.1%	6.9%	6.9%	
15	Richie Palacios TB	6.1%	7.0%	6.2%	6.0%	5.5%	6.2%	6.2%	6.0%	6.0%	
16	Steven Kwan CLE	5.2%	8.6%	5.5%	3.6%	4.2%	5.5%	5.5%	3.6%	5.2%	
17	Taylor Walls TB	4.1%	5.7%	4.2%	3.4%	3.4%	4.2%	4.2%	4.4%	4.3%	
18	Chandler Simpson TB	2.5%	5.6%	3.2%	1.4%	1.5%	3.2%	3.2%	1.4%	3.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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